

# CASE STUDY.

## HR Analytics Dashboard .

*How a diagnostic analysis of 1,470 IBM employee records uncovered the real reasons behind employee turnover — and turned them into a clear, actionable retention plan.*

**1,470**

Employees analyzed

**237**

Employees who left

**\$13.9M**

Estimated annual cost

**120**

Underpaid top performers

# A Company Was Losing Its Best People

Not because they were unhappy. Not because they underperformed. The real reason was hiding in the numbers.

A company with 1,470 employees came with one question: **why do we keep losing people, and what is it actually costing us?**

1,470

Total Employees

237

Employees Left

16.1%

Overall Attrition Rate

A 16.1% attrition rate is just a number — it doesn't say who is leaving, why, or what it costs. The real work was breaking that single number down into a pattern the business could actually act on.

## What Made This Diagnostic, Not Descriptive

- Every chart had to answer “why,” not just “what”
- Every assumption had to be tested against the data before being trusted
- Every finding had to translate into something a business could act on

# Following the Data, Layer by Layer

The analysis moved step by step — each finding raising the next question, rather than jumping straight to conclusions.

1

## Locate Where the Problem Lives

Broke the 16.1% down by department, role, and age. Sales had the highest departmental attrition (20.63%) — but inside it, the Sales Representative role alone hit 39.76%. Employees under 25 left at 39.1%, nearly four times the 35–44 age group.

2

## Compare Leavers to Stayers

Rather than guessing, every leading factor was tested directly: OverTime, travel frequency, manager-relationship satisfaction, and job involvement — each compared between those who left and those who stayed.

3

## Test the Obvious Assumption

The expectation was that employees with delayed promotions would be the flight risk. The data said the opposite — and that single test reshaped the entire recommendation.

4

## Find the Pay Gap Hiding in Plain Sight

Cross-referencing performance against job-level salary surfaced 120 high performers earning below their peers' average — a specific, fixable problem hiding behind a generic “improve culture” narrative.

5

## Put a Dollar Figure on All of It

Applied the SHRM 75%-of-salary replacement benchmark to every departure, then broken down by department — translating attrition from an HR concern into a financial one.

## Insight 1 — The Problem Was Never Evenly Spread

A single overall rate hides exactly where the risk is concentrated.

| Segment                     | Attrition Rate | Read                           |
|-----------------------------|----------------|--------------------------------|
| Sales (department)          | 20.63%         | Highest department overall     |
| Sales Representative (role) | 39.76%         | 4 in 10 employees leaving      |
| Under-25 employees          | 39.1%          | ~4× the 35–44 age group        |
| OverTime workers            | 30.5%          | 3× employees without overtime  |
| Low job involvement         | 33.73%         | vs. 9.03% for high involvement |

Frequent business travel and low satisfaction with one's manager also showed a clear link to departure — attrition reached **20.6%** at the lowest level of manager-relationship satisfaction. The pattern across all of these factors was consistent: **the problem wasn't who the employees were — it was how they were being managed.**

## Insight 2 — The Biggest Surprise in the Analysis

One assumption seemed obvious going in. The data said the opposite.

The expectation was simple: employees who haven't been promoted in a long time are the most likely to leave. It's the assumption almost every company makes.

| Time Since Last Promotion | Attrition Rate |
|---------------------------|----------------|
| 6+ years                  | 16.6%          |
| Recently promoted         | 28.57%         |

### WHY THIS MATTERS

Had this analysis stopped at intuition, the recommendation would have been to **accelerate promotions** — a costly fix for a problem that doesn't exist. The real risk window is the period right after a change (a promotion, a new manager, a new role), not the absence of one.

## Insight 3 — The Real Cost, and Where It Hides

The most valuable employees were also the most underpaid — and the most expensive department wasn't the one losing people fastest.

120

High performers paid below their job-level average

\$13.9M

Estimated annual attrition cost

\$7.5M

R&D's cost — despite a lower attrition rate

120 high-performing employees were earning **less than the average** for their own job level — not a company-wide pay issue, but a specific, fixable gap. This was the clearest signal that the retention problem went deeper than “improve engagement” — it was a measurable compensation issue with a measurable financial fix.

Using the SHRM benchmark — replacement cost at roughly **75% of annual salary** — total attrition cost came to an estimated **\$13.9M per year**. R&D didn't have the highest attrition rate, but because of its size and salary level, it carried the highest attrition cost at **\$7.5M** — proof that rate and cost are two different problems.

*Cost estimates use the SHRM 75%-of-annual-salary replacement benchmark applied to this dataset's actual compensation figures — directional planning estimates, not internal financial records.*

# What This Meant for the Business

The biggest opportunity wasn't in hiring. It was in keeping the people already there.

## The Conclusion

Three levers explained most of the risk, and all three were within the company's direct control:

- **Reduce reliance on OverTime** — the single strongest behavioral driver of departure
- **Improve the first-year experience** — for new hires and for employees with a new manager
- **Close the pay gap for high performers** — the 120-employee segment earning below their peers

## THE TAKEAWAY

None of these required a culture overhaul or a vague engagement initiative. Each one was a specific, measurable lever — with a specific, measurable financial return.

## What Was Delivered

- An interactive Power BI dashboard (7 pages, 20+ custom DAX measures)
- A written diagnostic report pairing every finding with a clear recommendation
- An executive storytelling presentation, delivered in English and Arabic
- A filterable list of the highest-risk employees, ready for direct HR follow-up